intuitive upgrades like adding a target network and implementing double Q-learning, so that is why we preferred to use value learning over policy methods in this book. However, in many cases, directly learning a policy function is more advantageous than a value function, such as for environments with a continuous action space, since we cannot create a DQN that returns an infinite number of Q values for each action.

11.2.3. Hierarchical reinforcement learning and the options framework

When a child learns to walk, they aren't thinking about which individual muscle fibers to activate and for how long, or when a businessperson is debating a business decision with colleagues, they aren't thinking about the individual sequences of sounds they need to make for the other people to understand their business strategy. Our actions exist at various levels of abstraction, from moving individual muscles up to grand schemes. This is just like noticing that a story is made up of individual letters, but those letters are composed into words that are composed into sentences and paragraphs and so on. The writer may be thinking of a general next scene in the story, and only once that's decided will they actually get to typing individual characters.

All the agents we've implemented in this book operate at the level of typing individual characters so to speak; they are incapable of thinking at a higher level. *Hierarchical reinforcement learning* is an approach to solving this problem, allowing agents to build up higher-level actions from lower ones. Rather than our Gridworld agent deciding one step at a time what to do, it might survey the board and decide on a higher-level sequence of actions. It might learn reusable sequences such as "move all the way up" or "move around obstacle" that can be implemented in a variety of game states.

The success of deep learning in reinforcement learning is due to its ability to represent complex high-dimensional states in a hierarchy of higher-level state representations. In hierarchical reinforcement learning, the goal is to extend this to representing states *and* actions hierarchically. One popular approach to this is called the *options framework*.